

Passive optical network planning in local access networks — an optimisation approach utilising genetic algorithms

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Optimal plant expansion in local access networks has strategic and financial importance for public telecommunications providers. Probably the most important future development in the area of digital communications will be the deployment of fibre optics by the telephone companies. This paper describes a genetic algorithm-based planning system developed to support the deployment of passive optical network (PON) technology in the local access network. Current planning practice is examined and the rationale behind the use of an optimisation method is explained. This paper presents a short introduction to genetic algorithms and outlines how they can be applied to PON planning. The system structure and optimisation framework associated with a genetic algorithm-based PON planning optimiser is discussed. The planning system developed by the authors can be used to determine the location, type, quantity and connection arrangements of PON components in networks. Results for some test problems are given; the optimiser is shown to produce low-cost solutions in an acceptable time.

1. Introduction

It has been estimated that the cost of installing fibre optic technology in the access network in a country the size of Great Britain will be in the order of fifteen billion pounds. Consequently, it is vital that network designs are optimised to minimise costs. It is anticipated that a significant proportion of the new fibre technology employed will be passive optical network (PON) systems.

This paper describes a genetic algorithm (GA) based planning optimisation system developed to support the deployment of PONs in the access network. The system takes an existing access duct infrastructure as an input and determines the location, type, quantity and connection arrangements of PON components so that expansion costs are minimised while still satisfying given customer demand.

2. Current planning

Access network expansion planning usually involves paper-based systems used in conjunction with manual or semi-automated calculations [1—3]. Plans are generally produced by the application of planning rules which ideally should be standardised and applied company-wide. With regard to the network planning problem under consideration there are two types of rules which may be applied:

- engineering rules, governing the technical feasibility of placement and interconnection of network elements — such rules are derived and validated in numerous tests by the equipment manufacturers,

- capacity management rules, relating to the provision of service capacity through the installation of new plant with consideration given to customer demand — these rules are empirically derived by planners from their experience gained when planning the local network.

While engineering rules can be regarded as accepted, *de facto* standards, capacity management rules suffer from the following limitations:

- the application of a concise set of rules is unlikely to produce optimal results for all conceivable networks,
- capacity management rules have not yet been validated and shown to produce lowest cost plant expansion plans.

In order that capacity management rules may be applied by a human planner, the set of rules must be kept as small and as simple as possible so that they are easy to comprehend. For network planning, which is generally recognised to be a very complex process, it is unlikely that such a rule set will produce minimal cost plans for all conceivable networks.

As there is currently no known method which produces optimal expansion plans, validation of the effectiveness of capacity management rules can only be performed empirically. However, since network planning is a complex operation, the time required for a planner to complete a new expansion plan is considerable.

Therefore, a process is desirable in which the calculation of costs is computer-aided so that a large number of plans can be evaluated rapidly. In addition, optimisation procedures must be found which can intelligently search the colossal number of alternative plans for all possible networks.

Computer-based optimisation methods have already been successfully applied to other network planning problems. It has been demonstrated [4, 5] that significant savings can be made when compared to the previous manual planning method employed. In particular, an optimisation-based planning approach has been used to expand conventional copper networks [5], resulting in an average saving of 10%.

3. Computer-aided planning

Access network expansion planning is a multi-faceted problem. The number of real-world planning considerations, options and network structures makes the problem extremely challenging. The solution space for network planning is excessively large and difficult to search because of its structure.

In order to make the problem solvable by computational optimisation methods, the complexity of the problem formulation must be reduced. This can be achieved in two principal ways:

- the overall problem can be divided into sub-problems which are solved individually,
- simplifying assumptions can be introduced.

The choices relating to the method of decomposition and which assumptions to make are decisive factors for the success of an optimisation approach to practical planning.

Decomposition of the overall problem into sub-problems is a common practice in network planning. When decomposing a problem care must be taken to minimise inter-dependencies. As this task is often difficult to achieve multiple iterations through the sub-problems are usually undertaken. Simplifying assumptions concern the factors taken into consideration when evaluating a plan, the number of alternative planning options or the properties of a particular network. In order to produce plans which can actually be implemented in the field only assumptions which retain the dominant problem characteristics are permissible.

Given the limitations already discussed, the aim of access network expansion planning must be redefined to be the identification of near-optimal instead of optimal plans. In practice, this means that solutions are acceptable if they improve on current planning schemes.

3.1 PONs and access networks

An access network is the part of the public telecommunications network which connects customers to the level at which switching takes place, i.e. to the central exchange. A network of nodes connected by duct routes determines the structure of an access network.

The main components of a PON are optical fibre strands and optical light splitters (see Fig 1).

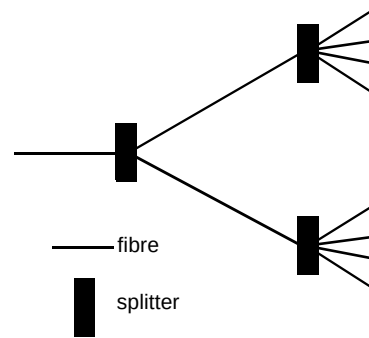


Fig 1 PON schematic.

A splitter is capable of evenly splitting an incoming optical signal into two or more signals so that all outputs transmit the same information as that of the input but at a reduced light level. Splitters are passive components which do not require a power supply to operate.

The system consists of the exchange unit, a set of splitters and the customer terminating equipment. All splitter components and their interconnecting fibre strands are part of the same PON. As splitters transmit the complete information which enters their inputs to all of their outputs, the optical information transmitted is identical at all points in a PON. The allocation of data to specific customers is managed by the exchange and terminating equipment. A PON generally contains splitters in series, so that a signal may be split more than once. Exactly how many splitters in series are permissible and in which configuration depends on the specification of the particular equipment being used. The planning system described in this paper is based on BT's OTIAN equipment.

3.2 PON optimisation

A specific planning problem is defined by an access network structure in terms of nodes and ducts and by customer demand associated with network terminating nodes. The principal aim of access network expansion planning is to determine PON layouts for a given network such that all given customer demand is satisfied, at

minimum cost. A solution to the problem details information such as the location, type, quantity and connection arrangements of PON components. Demand may be satisfied by a single or multiple PONs.

More formally, the optimisation problem is defined by a set of decision variables, an objective function to be minimised and a set of constraints associated with decision variables. Decision variables are the location, type, quantity and connection arrangements of PON components. The sum of all decision variables defines a solution to the problem. The objective function, incorporating the decision variables, allows the evaluation and comparison of alternative solutions. In this case, the objective function determines the actual cost of PON installations. There are two types of constraints on solutions. The first set of constraints ensures that all customer demand is satisfied by a solution, whereas the second set of constraints ensures the technical feasibility of a solution.

Expansion planning can be visualised as a mapping process in which one or more PONs are mapped on to a given access network structure, as depicted in Fig 2. In this process, splitters must be assigned to network nodes and fibre cables to ducts. Given the range of PON-type configurations which arise from different splitter combinations and the potentially large number of network nodes, the total number of mappings is enormous.

Access network expansion belongs to the class of combinatorial optimisation problems. It has been recognised that network planning problems are generally difficult to solve. This difficulty arises from a combination of the size and structure of the search space. The size of the search space for an access network expansion problem of moderate dimensions is already so big that even the fastest computer would need far longer than the estimated age of the universe to search it exhaustively. The objective function and constraints of the optimisation problem are nonlinear and decision variables have a discrete range. In addition, the problem cannot easily be converted to a form to which a known efficient solution procedure can be applied.

The sum of these problem features makes the application of standard optimisation methods difficult if not impossible. Furthermore, in such a situation a near optimal solution is the best that can be expected and consequently heuristic methods are often the only option.

Genetic algorithms (GA) are heuristic search algorithms which have already successfully been used to solve other difficult combinatorial optimisation problems. An important consideration is that GAs do not require any features such

as linear or differentiable objective functions and are therefore applicable to problems with arbitrary structures.

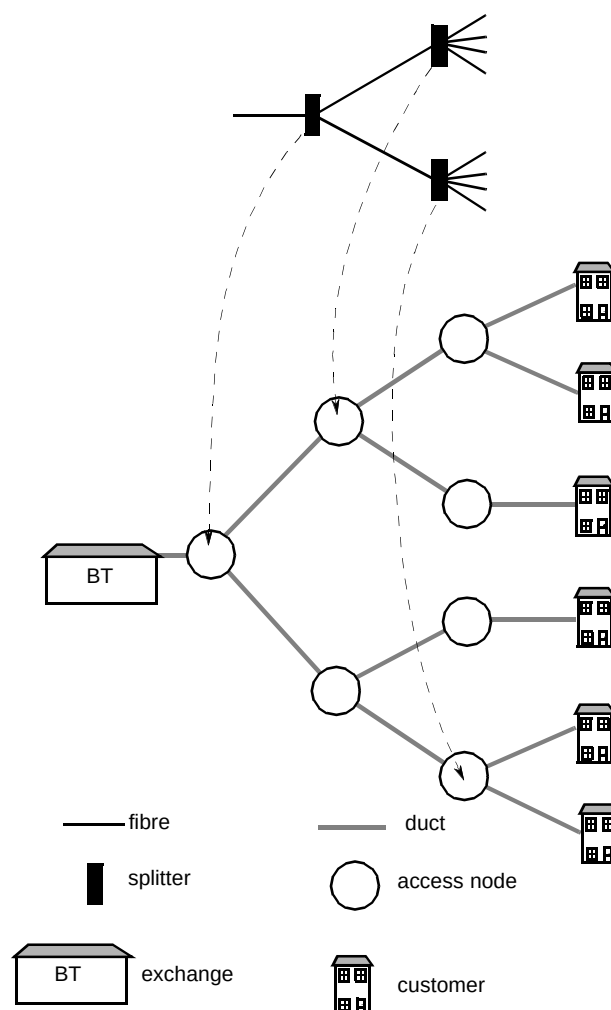


Fig 2 PON mapping on to an access network.

4. Genetic algorithms

The basic rationale behind GAs is rooted in the observation that, considering the vast amount of genetically possible life forms, plants and animals have evolved faster than randomly possible. GAs emulate evolution, the driving force behind developments in nature, in a number of ways.

Similar to natural species, which evolve in populations among which cross-fertilisation takes place, GAs simultaneously evolve multiple solutions to a given problem. Instead of working on a single solution, as is the case for most optimisation methods, GAs work in parallel on a population of solutions.

As solutions evolve, the mechanism of survival of the fittest is employed to determine which solutions to keep in the population during an optimisation and which to discard. Each solution is evaluated and promoted into the next generation of solutions according to its fitness for the specific purpose. The selection is performed stochastically and the probability of a solution surviving into the next generation is proportional to its fitness. This process allows less fit solutions to occasionally be selected and therefore supports genetic diversity in a population.

Decision variables are generally encoded for the application of GAs. The individual variables are called genes, the sum of all genes which is the solution, is called a chromosome. The most frequently used encoding is binary, and consequently in most cases chromosomes are represented by binary strings. However, this form of representation is not the only one possible. Chromosomes can be represented in a number of forms suitable for a given problem.

Solutions are evolved by genetic operators. For a canonical GA the operators are crossover and mutation. Crossover is based on the principle of genetic mixing, during which genes of one chromosome are recombined with genes of another chromosome. Mutation on the other hand is an operator which works on a single chromosome making only small changes to the genetic information.

GAs are a family of search algorithms which can vary with regard to representation of chromosomes, operators and selection mechanisms. In fact, the representation and operators for this work are non-standard; a set of special operators have been developed to solve the PON expansion problem. Genetic algorithms originate from the work by Holland [6]; recently they have been presented with new applications by, for example, Goldberg [7] and Davis [8].

5. System design

In this section, the overall architecture of a GA-based expansion planning tool is outlined, the top-level diagram is shown in Fig 3.

The first step in an optimisation process is the identification of a particular access network which must be expanded. The access network can be loaded from network databases, displayed and, if necessary, modified by the visualisation module. Alternatively, the network can be created graphically by the visualisation module. After the target access network is completely specified, it is submitted as input data to the optimisation module. Input data must specify the access network structure and customer demands. The optimisation module in turn produces an output which specifies the location, type, quantity and connection arrangements of PON components in relation to the given access network. This output is ultimately fed back into the visualisation module which displays the outcome of the optimisation process. A planner will be able to inspect, at a macro level, an optimisation solution as a mapping of one or more PONs on to the access network structure. In addition, details relating to the interconnections of components and their costs can also be displayed.

Figure 4 shows the structure of the optimisation module and illustrates how GAs can be applied to network planning. It is evident that solution evaluation and constraint checking require data about a specific network. This data located in the network model is entered via the visualisation module and transferred to the optimisation module. As GAs are general-purpose search algorithms, they must be coupled to a specific problem via an encoded representation of a solution and an evaluation function. The coupling of the problem domain to the GA module is managed by the problem interface module. In the case of expansion planning, a solution is a plan; the template generator creates a plan template (an empty solution framework) for a specific network which is then transferred to the GA module prior to an optimisation run. The encoded plan

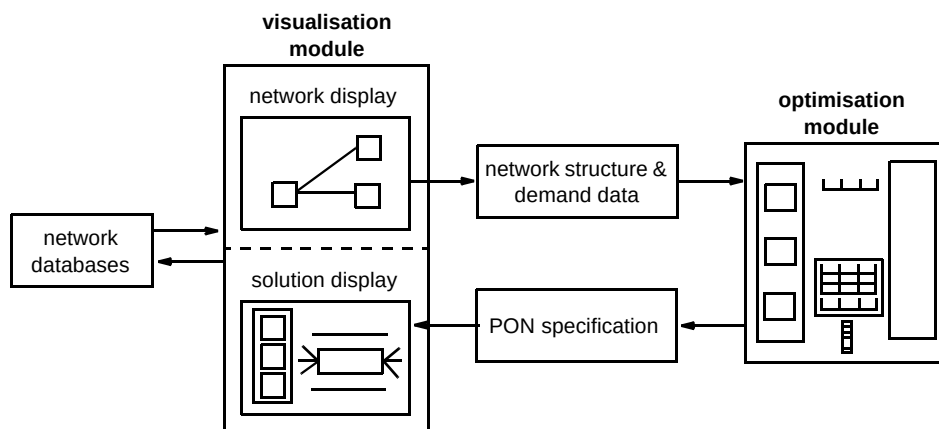


Fig 3 Overall system architecture.

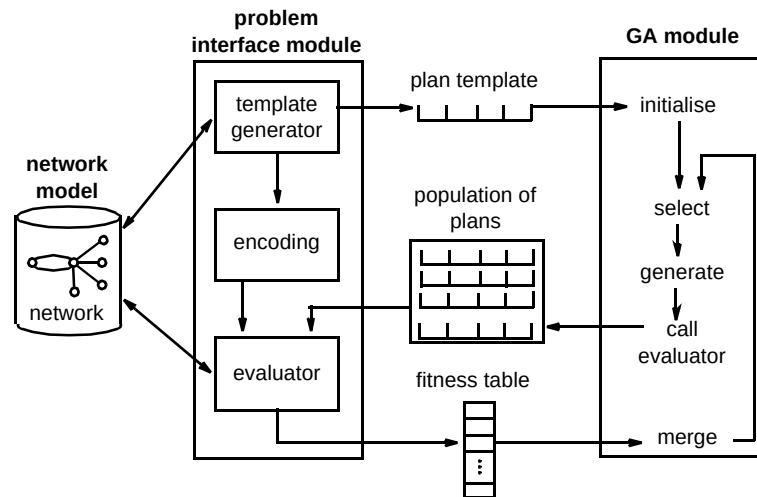


Fig 4 Structure of the optimisation module.

template remains static during the optimisation process. The solution encoding is stored as it is required as a reference point during the evaluation of a solution. The structure and size of a plan template depends directly on the structure of the access network to be expanded as stored in the network model.

The GA module iterates through a loop:

- selecting solutions according to their fitness,
- generating new solutions by means of operators,
- evaluating the population of solutions,
- merging new solutions into the population of existing solutions.

As the evaluation of solutions is problem specific, it is performed by the evaluator external to the GA module. To acquire a fitness value for each solution the evaluator interacts with the network model.

6. Results

The quality of optimisation results can be assessed in two ways. Firstly, the optimiser can be judged for its ability to solve the expansion problem as formulated in the evaluation function and system constraints. Secondly, the optimisation results can be evaluated regarding their quality for practical PON planning. The two considerations are distinct because of the decompositions and assumptions made in the formulation of the optimisation problem. As expansion planning is a very challenging task to undertake, for which no procedures are known to determine optimal results, an objective assessment is difficult.

optimum solutions are believed to be known. First tests were conducted with a standard GA having operators which are not problem specific. The standard operators performed unsatisfactorily; no parameter settings could be found which produced results better than 10% above the best known cost. New research into operators tailored for the expansion problem, however, led to the production of excellent results. By using the new operators parameter settings can be found which generally find the optimum solution for the test networks.

Figure 5 shows the course of two typical optimisation runs for the optimiser with standard operators and with the new operators on a network with 25 nodes. The best solutions obtained during runs are plotted against the number of evaluated solutions; the best-known cost is normalised to 1. The new operators make the solution converge at a faster rate towards the optimum and more importantly, lead ultimately to the best-known solution. On average the optimum is found after 10 000 solutions have been evaluated. On a 486 PC running at 66 MHz, this corresponds to an elapsed time of 20 seconds.

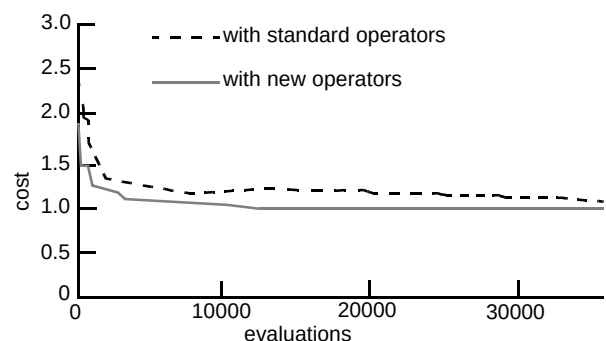


Fig 5 Typical optimisation run for a 25-node network.

114 Initially the numerical performance of the optimiser has been tested on networks of moderate size for which the

7. Discussion and conclusions

A GA-based optimisation system for PON expansion planning in access networks has been designed, implemented and tested. To the knowledge of the authors this is the first paper dealing with optimisation methods for PON planning.

It was found that GAs with standard representation and operators did not produce satisfactory results for this problem. Operators tailored for the PON expansion problem had to be developed. These new operators were shown to produce excellent results. While in this paper computational results were obtained only for test networks of moderate size, it is anticipated that the approach can successfully be applied to networks of larger dimensions. First experiments with networks of 60 nodes show that the time to compute high-quality solutions is in the order of ten minutes.

It should be noted that improvement in PON expansion planning can be achieved not only by cost reductions but also by a decrease in the time taken to perform planning. As tests were performed on a low-cost PC, computation time could easily be reduced further by utilising a more powerful platform.

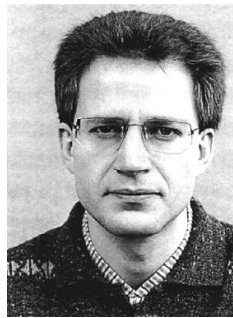
In conclusion, the quality of optimisation results for real PON planning depends on the suitability of decompositions and assumptions present in the formulation of the optimisation problem. As the optimiser was developed in collaboration with BT planners it is believed that the dominant components of the problem have been retained in the formulation. The validity of derived solutions may be tested by comparison with real networks. Such tests are planned for the future.

Acknowledgements

This project was conducted in collaboration with BT personnel and, in particular, we would like to thank Stephen Brewis for his input throughout the project and Don Asumu for the design and implementation of the visualisation module. We would also like to thank Professor Hugh Ryan and Christian Woeste at the University of Sunderland for many helpful discussions.

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